II-Probabilistic Approaches Independence and Graphical Models

Foundations of Neuro-Symbolic AI

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Outline

- Independence of variables
- Conditional independence
- Decomposition schemes
- Bayesian networks
- Markov networks

How to beat the curse of dimensions?

Probability distribution of factored systems with states $\bigtimes_{k\in[d]}[m_k]$ has

$$\left(\prod_{k\in[d]} m_k\right) - 1$$

degrees of freedom (coordinates to specify and store).

Mitigation: Tensor network decompositions

Idea

Exploit the "dependence" structure of variables to find decompositions of probability distributions.

Example: At the dentist

Consider the joint distribution of the variables X_0 (Toothache), X_1 (Cavity), X_0 (Catch), with a variable X_3, denoting whether there is a Cloud outside the dentists lab.

- We now have an order-4 probability tensor $\mathbb{P}\left[X_0, X_1, X_2, X_3\right]$
- Intuitively, knowing Cloud should not affect the probability of having a cavity, so why should we care?

To be more precise let us consider the distribution:

$$\mathbb{P}\left[X_0, X_1, X_2, X_3\right] = \begin{matrix} \begin{pmatrix} 0.288 & 0.004 \\ 0.032 & 0.006 \end{pmatrix} \begin{pmatrix} 0.077 & 0.036 \\ 0.008 & 0.054 \end{pmatrix} \\ \\ \begin{pmatrix} 0.288 & 0.004 \\ 0.032 & 0.006 \end{pmatrix} \begin{pmatrix} 0.077 & 0.036 \\ 0.008 & 0.054 \end{pmatrix} \end{matrix} [X_0, X_1, X_2, X_3]$$

Such redundancies are captured by the concept of independence.

Formal definition of independence

Definition (Independence)

Given a joint distribution $\mathbb{P}[X_{\mathcal{V}}]$ and two disjoint subsets $A, B \subset \mathcal{V}$, we say that the variables X_A are independent from the variables X_B if the marginal distributions $\mathbb{P}[X_{A\cup B}]$, $\mathbb{P}[X_A]$ and $\mathbb{P}[X_B]$ satisfy for any index tuples x_A, x_B

$$\mathbb{P}[X_A = x_A, X_B = x_B] = \mathbb{P}[X_A = x_A] \cdot \mathbb{P}[X_B = x_B] \ .$$

We denote in this case $(X_A \perp X_B)$.

Interpretation: Knowing the value of variables B (and conditioning on them) does not change the distribution of variables A.

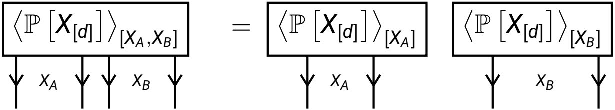
Contraction criterion for independence

Theorem

If and only if the X_A are independent from the variables X_B with respect to a distribution $\mathbb{P}[X_{\mathcal{V}}]$ then:

$$\left\langle \mathbb{P}\left[X_{[d]}\right]\right\rangle_{[X_A, X_B]} = \left\langle \mathbb{P}\left[X_{[d]}\right]\right\rangle_{[X_A]} \otimes \left\langle \mathbb{P}\left[X_{[d]}\right]\right\rangle_{[X_B]} .$$

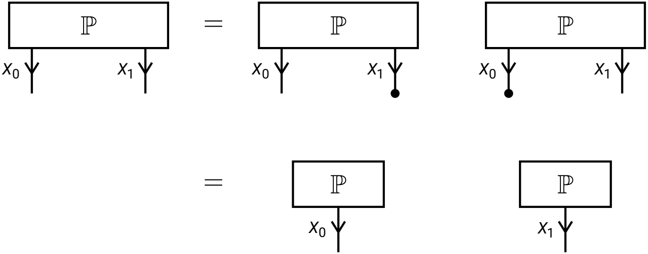
In a tensor network diagram this is the decomposition:



Decomposition into marginal distributions

Independence allows the decomposition of proability tensors into tensor products.

Example: The distribution of two independent variables decomposes as:



Storage advantage

Instead of storing $m_0 \cdot m_1$ coordinates, we can store $\mathbb{P}$ in the case of independence with $m_0 + m_1$ demand.

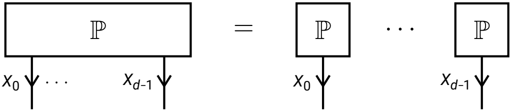
Decomposition of pairwise independent variables

Imagine now that we have a distribution $\mathbb{P}\left[X_{[d]}\right]$, such that for any $k, \tilde{k} \in [d]$

$$\left(X_k \perp X_{\tilde{k}}\right) \ .$$

We then have the decomposition:

$$\mathbb{P}\left[X_{[d]}\right] = \bigotimes_{k\in[d]} \mathbb{P}\left[X_k\right]$$



Exponential to linear storage demand

Instead of storing $\prod_{k\in[d]} m_k$ coordinates, we can store $\mathbb{P}$ using $\sum_{k\in[d]} m_k$ coordinates of the marginal distributions.

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Assumption of independence

Independence is a very restrictive assumption:

- We need dependence of effects on causes for machine learning to work.
- Independence covers only the most extreme form of tensor network decompositions.

Example: In the dentist example (without the Cloud variable), no variables are independent from each other.

Are there more common properties corresponding to more generic tensor decompositions?

Formal definition of Conditional Independencies

Definition (Conditional Independence)

Given a joint distribution of variables X_A, X_B and X_C, we say that X_A and X_B are independent given X_C if for any index tuple (x_A, x_B, x_C)

$$\begin{aligned}\mathbb{P}\left[X_A = x_A, X_B = x_B \middle| X_C = x_C\right] = &\ \mathbb{P}\left[X_A = x_A \middle| X_C = x_C\right] \\ &\cdot \mathbb{P}\left[X_B = x_B \middle| X_C = x_C\right] ,\end{aligned}$$

We denote this situation by

$$(X_A \perp X_B) \mid X_C .$$

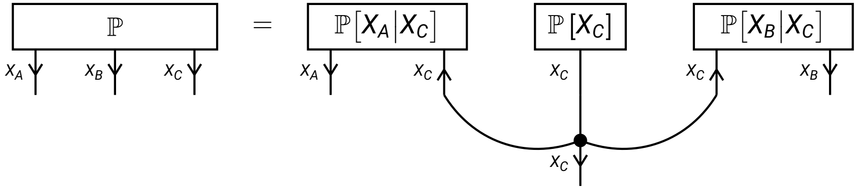
Interpretation: Conditional independence means that X_A and X_B are independent with respect to any slice $\mathbb{P}\left[X_A, X_B \middle| X_C = x_C\right]$.

Conditional independence as a contraction equation

We have $(X_A \perp X_B) \mid X_C$ if and only if

$$\mathbb{P}[X_A, X_B | X_C] = \left\langle \mathbb{P}[X_A | X_C], \mathbb{P}[X_B | X_C] \right\rangle_{[X_A, X_B, X_C]} .$$

We depict conditional independence by tensor network decompositions:



Example: Being at a dentist

Task: Let us investigate, whether Toothache (X_0) and Catch (X_2) are independent given Cavity (X_1), with respect to the distribution:

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

For $X_1 = 0$:

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For $X_1 = 0$:

$$\mathbb{P}\big[X_0, X_2 \big| X_1 = 0\big] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2]$$

Example: Being at a dentist

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For $X_1 = 0$: Independent ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} [X_2]$$

Example: Being at a dentist

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For $X_1 = 1$:

$$\mathbb{P}\left[X_0, X_2 \middle| X_1 = 1\right] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2]$$

Example: Being at a dentist

Task: Let us investigate, whether Toothache (X_0) and Catch (X_2) are independent given Cavity (X_1), with respect to the distribution:

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For $X_1 = 1$: Independent ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.1 \\ 0.9 \end{pmatrix} [X_2]$$

Example: Being at a dentist

Task: Let us investigate, whether Toothache (X_0) and Catch (X_2) are independent given Cavity (X_1), with respect to the distribution:

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

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Conclusion: The conditional independence holds, that is $(X_0 \perp X_2) \mid X_1$.

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Chain Rule: Decomposing Probabilities

Theorem (Chain Rule)

For any joint probability distribution $\mathbb{P}\left[X_{[d]}\right]$ we have

$$\mathbb{P}\left[X_{[d]}\right] = \left\langle \{\mathbb{P}\left[X_k \middle| X_{[k]}\right] \,:\, k \in [d]\} \right\rangle_{\left[X_{[d]}\right]} .$$

Proof sketch: We apply the Bayes theorem to get

$$\mathbb{P}\left[X_{[d]}\right] = \left\langle \mathbb{P}\left[X_0\right], \mathbb{P}\left[X_1, \cdots, X_{d-1} \middle| X_0\right] \right\rangle_{\left[X_{[d]}\right]} .$$

and further apply the Bayes theorem iteratively on the conditioned distribution.

Problem: The tensors $\mathbb{P}\left[X_k \middle| X_{[k]}\right]$ are of order $d+1$!

Can we reduce the order based on independence assumptions?

Simplification of conditional distributions

Theorem

When $(X_A \perp X_B) \mid X_C$ *then*

$$\mathbb{P}\left[X_A \middle| X_C, X_B\right] = \mathbb{P}\left[X_A \middle| X_C\right] \otimes \mathbb{I}\left[X_B\right] .$$

Proof sketch: We have by Bayes theorem

$$\mathbb{P}\left[X_A, X_B \middle| X_C\right] = \left\langle \mathbb{P}\left[X_A \middle| X_C, X_B\right], \mathbb{P}\left[X_B \middle| X_C\right] \right\rangle_{[X_A, X_B, X_C]}$$

and compare with the conditional independence equation

$$\mathbb{P}\left[X_A, X_B \middle| X_C\right] = \left\langle \mathbb{P}\left[X_A \middle| X_C\right], \mathbb{P}\left[X_B \middle| X_C\right] \right\rangle_{[X_A, X_B, X_C]} .$$

Usage for decompositions: We can drop some conditioned variables and effectively reduce the tensor order in the tensors $\mathbb{P}\left[X_k \middle| X_{[k]}\right]$.

Example: Markov Chains

Let us model the weather in a town on all days of a year by variables $X_{[365]}$ and assume that for each $k \in [365]$

- Three states $x_k = 0$ (sunny), $x_k = 1$ (cloudy), $x_k = 2$ (rainy)
- The weather on the day $k+1$ conditioned on the weather on the day k is

$$\mathbb{P}\big[X_{k+1} \big| X_k\big] = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.35 & 0.40 & 0.25 \\ 0.10 & 0.40 & 0.50 \end{pmatrix} [X_{k+1}, X_k]$$

- The weather on the day $k+1$ does not depend on the weather before day k given the weather before day k:

$$\big(X_{k+1} \perp X_{[k]}\big) \mid X_k$$

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- The weather on the day $k+1$ does not depend on the weather before day k given the weather before day k:

$$\left(X_{k+1} \perp X_{[k]}\right) \mid X_k$$

This is an example of a Markov chain: Variables depend only on their predecessors.

Example: Markov Chains

Theorem (Markov Chain)

Let there be a set of variables X_k where $k \in [d]$. When X_k is independent of $X_{[k-1]}$ given X_{k-1} (the Markov Property), then

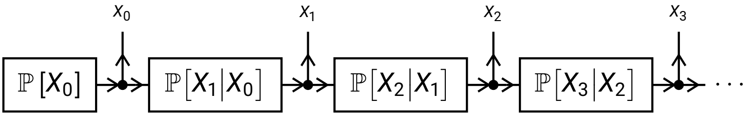
$$\mathbb{P}\left[X_{[d]}\right] = \left\langle \mathbb{P}\left[X_k \middle| X_{k-1}\right] \,:\, k \in [d] \right\rangle_{\left[X_{[d]}\right]} .$$

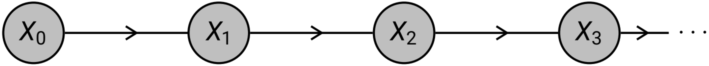
Proof sketch: We use the chain decomposition and exploit the independence assumptions to get

$$\mathbb{P}\left[X_k \middle| X_{[k]}\right] = \mathbb{P}\left[X_k \middle| X_{k-1}\right] \otimes \mathbb{I}\left[X_{[k-1]}\right] .$$

Tensor network decompositions of markov chains

We depict this by





Can we find a more generic correspondence between probability decompositions and conditional independence statements?

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Independence assumptions of Bayesian networks

Towards decomposing a distribution in a more generic way take the steps:

- We choose an order and enumerate the variables by $[d]$.
- To each $k \in [d]$ we choose a set $\mathrm{Pa}(k)$ of parent nodes in $[k] = \{0, \cdots, k-1\}$ such that

$$\left(X_k \perp X_{[k]\setminus \mathrm{Pa}(k)}\right) \mid X_{\mathrm{Pa}(k)}$$

Example: Markov chains are the special case of $\mathrm{Pa}(k) = \{k-1\}$.

Factorization into Bayesian networks

Theorem

When for each $k \in [d]$

$$\left(X_k \perp X_{[k]\backslash \mathrm{Pa}(k)}\right) \mid X_{\mathrm{Pa}(k)}$$

then

$$\mathbb{P}\left[X_{[d]}\right] = \left\langle \left\{ \mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right] \,:\, k \in [d] \right\} \right\rangle_{\left[X_{[d]}\right]} .$$

Proof sketch: As in the proof of the Markov chain factorization theorem, we use the chain factorization and exploit the conditional independences to show that

$$\mathbb{P}\left[X_k \middle| X_{[k]}\right] = \mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right] \otimes \mathbb{I}\left[X_{[k]\backslash \mathrm{Pa}(k)}\right] .$$

Example: At the Dentist

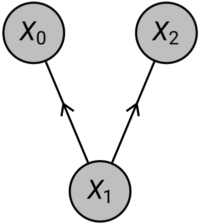
- Choose the order X_1, X_0, X_2
- For both X_0 and X_2 we take X_1 as single parent, that is

$$\mathrm{Pa}(0) = \{1\} \quad \text{and} \quad \mathrm{Pa}(2) = \{1\}$$

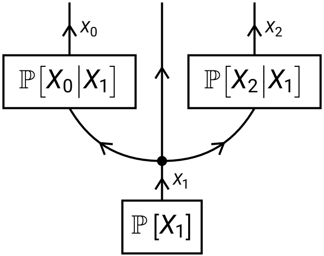
  and have the for the Bayesian network necessary conditional independencies (shown before)

$$(X_0 \perp X_2) \mid X_1 \quad \text{and} \quad (X_2 \perp X_0) \mid X_1$$

The corresponding hypergraph is:



$\mathbb{P}[X_0, X_1, X_2] =$



Example: Hidden Markov Models

Extend the weather example: We observe the weather X_k, but not the hidden driving variable:

- Atmospheric pressure Y_k taking states $y_k = 0$ (high pressure) or $y_k = 1$ (low pressure)
- The pressure at a day $k + 1$ is independent of the pressure at days $[k]$ given the pressure at day k:

$$\left(Y_{k+1} \perp Y_{[k]}\right) \mid Y_k$$

- The weather at day k depends only on the pressure:

$$\left(X_k \perp Y_{[k]}, X_{[k]}\right) \mid Y_k$$

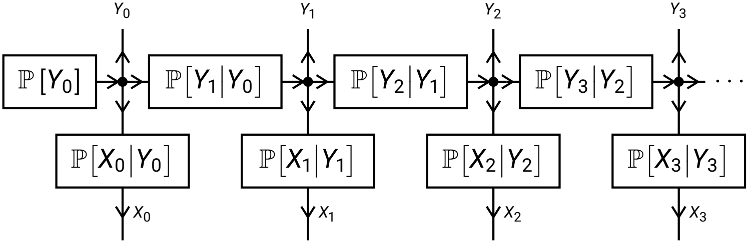
This is an example of Hidden Markov Models extending Markov Chains by limited observation X_k of the variables Y_k.

Example: Hidden Markov Models

The independence assumptions of a Hidden Markov Model are for any $k \in [d]$

- Y_{k+1} is independent of $Y_{[k]}$ and $X_{[k]}$ given Y_k
- X_k is independent of all other variables given Y_k

The independence assumptions are exploited in the decomposition



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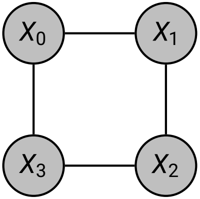
Sometimes there is no clear order of the variables

Example: Consider four students enumerated by $k \in [4]$ and

- Boolean variables X_k whether they know the solution to a problem.
- They sit next to each other in a circle and sometimes pass the solution to their neighbor.



We have the independencies:

$$(X_0 \perp X_2) \mid X_1, X_3 \quad \text{and} \quad (X_1 \perp X_3) \mid X_0, X_2$$

But: Any order of the student variables in a Bayesian network would at most capture one of these two independencies.

Representation as a Markov network

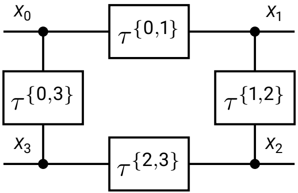
Let us make the neighboring students example more precise:

- For each pair of neighbors, we "increase" by a factor 2 the probability that both either know or do not know the solution.
- We define matrices:

$$\tau^{\{0,1\}}[X_0, X_1] = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} [X_0, X_1]$$

- We then model the distribution by a tensor network:

$$\mathbb{P}\left[X_{[4]}\right] = \frac{1}{\left\langle \tau^{\{0,1\}}, \tau^{\{1,2\}}, \tau^{\{2,3\}}, \tau^{\{0,3\}} \right\rangle_{[\varnothing]}} \cdot$$



Different to Bayesian networks: Need to normalize the distribution

$$\mathbb{P}\left[X_{[4]}\right] = \left\langle \tau^{\{0,1\}}, \tau^{\{1,2\}}, \tau^{\{2,3\}}, \tau^{\{0,3\}} \right\rangle_{[X_{[4]}|\varnothing]}$$

Undirected Graphical Models: Markov Networks

Definition (Markov Networks)

Let $\{\tau^{\mathrm{e}}[X_{\mathrm{e}}] : \mathrm{e} \in \mathcal{E}\}$ be a Tensor Network on a hypergraph $\mathcal{G}$. The associated Markov Network is the probability distribution of $\{X_v : v \in \mathcal{V}\}$ defined by

$$\mathbb{P}[X_{\mathcal{V}}] = \frac{\langle\{\tau^{\mathrm{e}}[X_{\mathrm{e}}] : \mathrm{e} \in \mathcal{E}\}\rangle_{[X_{\mathcal{V}}]}}{\langle\{\tau^{\mathrm{e}}[X_{\mathrm{e}}] : \mathrm{e} \in \mathcal{E}\}\rangle_{[\varnothing]}} = \langle\{\tau^{\mathrm{e}}[X_{\mathrm{e}}] : \mathrm{e} \in \mathcal{E}\}\rangle_{[X_{\mathcal{V}}|\varnothing]} .$$

We call the denominator

$$\mathcal{Z}(\mathcal{G}) = \langle\{\tau^{\mathrm{e}}[X_{\mathrm{e}}] : \mathrm{e} \in \mathcal{E}\}\rangle_{[\varnothing]}$$

the partition function of the Markov Network.

Markov network independencies

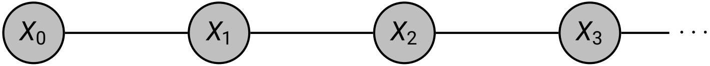
Which conditional independencies hold in a Markov network on a hypergraph $\mathcal{G}$?

Definition

For any hypergraph $\mathcal{G} = (\mathcal{E}, \mathcal{V})$ we define:

- A path through $\mathcal{G}$ is a selection of nodes $[d] \subset \mathcal{V}$ such that for any $k \in [d-1]$ we find a hyperedge $e \in \mathcal{E}$ with $k, k+1 \in e$.
- Given disjoint subsets $A, B, C \subset \mathcal{V}$ we say that C separates A from B if any path starting at a node in A and ending at a node in B goes through C.

Example: In a Markov chain graph, for any $k \in [d]$ any subset $A \subset [k]$ and subset $B \subset [d] \backslash [k+1]$ is separated by $C = \{k\}$.



Markov network independencies

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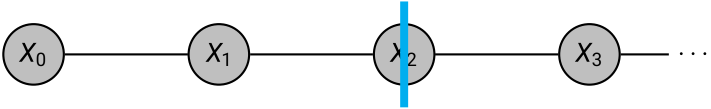
- Given disjoint subsets $A, B, C \subset \mathcal{V}$ we say that C separates A from B if any path starting at a node in A and ending at a node in B goes through C.

Example: In a Markov chain graph, for any $k \in [d]$ any subset $A \subset [k]$ and subset $B \subset [d] \backslash [k+1]$ is separated by $C = \{k\}$. For $k = 2$:

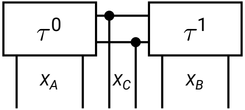


Markov network independencies

Correspondence of separation and independence: When subsets C separates A from B, then we have

$$(X_A \perp X_B) \mid X_C \,.$$

This can be shown based on the disentanglement of networks (see lecture on CSPs):



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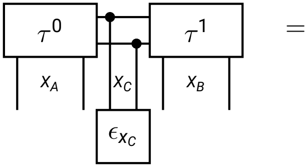

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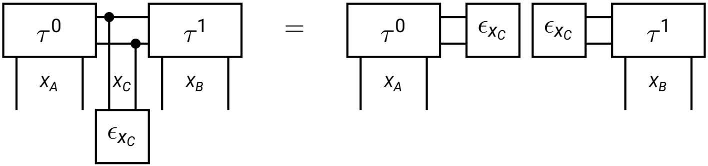

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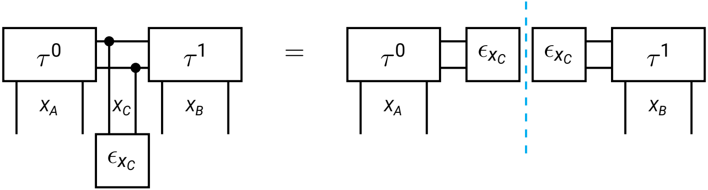

Conclusion: Given an assignment to X_C, the variables X_A must be independent of X_B.

Markov network indepencencies

The other direction is true for positive distributions: We need to assume for any $x_{[d]}$

$$\mathbb{P}\left[X_{[d]} = x_{[d]}\right] \neq 0\,.$$

Theorem (Hammersley Clifford)

A positive distribution $\mathbb{P}\left[X_{\mathcal{V}}\right]$ has a representation as a Markov network on a hypergraph $\mathcal{G}$, if and only if for any disjoint subsets $A, B, C \subset \mathcal{V}$ for which C separates A from B

$$(X_A \perp X_B) \,|\, X_C\,.$$

Proof: Very technical, see the book [Koller, Friedman: Probabilistic Graphical Models 2009] and alternatively using the contraction formalism the paper [Goessmann et al: A tensor network formalism for neuro-symbolic AI].

Example: Recommending a student

Consider a distribution of the variables:

- Intelligence of a student: X_I
- SAT score of the student: X_S
- Difficulty of a test: X_D
- Grade of the test: X_G
- Recommendation by the teacher: X_L

Let us assume the independencies:

- "The SAT score depends only on the student's intelligence":

$$\left(X_S \perp X_{\{D,G,L\}}\right) \mid X_I$$

- "The recommendation letter depends only on the grade":

$$\left(X_L \perp X_{\{D,I,S\}}\right) \mid X_G$$

The corresponding hypergraph: